
Solvent-Hydrolysis-Driven Engineering of Ordered Single Quantum Well 2D Perovskites

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Solvent-hydrolysis-driven engineering of ordered single quantum well 2D perovskites

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ABSTRACT Single quantum well (single-QW) two-dimensional (2D) perovskites are set to transform optoelectronic devices due to their stability and superior properties. However, solution-processed 2D perovskites often form disordered multiple-QW structures, leading to inconsistent performance. Here, we present a solvent-hydrolysis-driven method for controlling crystallization kinetics, resulting in highly ordered single-QW 2D perovskite films. Dimethylamine (DMA), formed from the hydrolysis of N,N-dimethylformamide (DMF), acts as a key mediator, preventing cluster aggregation and ensuring uniform colloidal distribution. This process avoids a heterogeneous intermediate phase, thereby fostering the formation of a homogeneous (DMA,MA)PbI₃ phase, which is crucial for the development of single-QW films. The resulting photodetector demonstrates outstanding performance, with a responsivity of 1153 mA/W and a detectivity of 6.98×10^{12} Jones, along with excellent photostability under ambient conditions. These attributes make it ideal for photoelectric imaging sensors and large-scale integration. Our findings provide a scalable, solution-processed strategy for high-performance 2D perovskite materials, opening up new possibilities for advanced optoelectronic applications.

Keywords: 2D perovskite, Single-quantum-well, Solvent hydrolysis, Optoelectronic device, Stability

INTRODUCTION

Metal halide perovskites have attracted significant interest for optoelectronic applications due to their exceptional properties, including high light absorption coefficients, long carrier lifetimes, high mobility, and cost-effective synthesis^[1-3]. However, their application is limited by poor long-term stability, primarily due to their soft lattice, which makes them susceptible to environmental factors such as heat, humidity, and

oxygen^[4,5]. Two-dimensional (2D) perovskites, derived from their three-dimensional (3D) counterparts by incorporating organic cations, offer enhanced stability and adjustable optoelectronic properties, making them promising for future devices^[6]. These materials follow the general formula $A'_m A_{n-1} B_n X_{3n+1}$, where A' is a long-chain organic cation, A is a monovalent cation, B is a divalent metal cation, and X is a halide anion, with $m = 2$ in Ruddlesden-Popper (RP) phases and $m = 1$ in Dion-Jacobson (DJ) phases^[7,8]. They consist of n -layer inorganic $[BX_6]^{4-}$ octahedral slabs separated by organic ligands, creating a natural quantum well (QW) due to the different dielectric constants between the layers^[9].

Despite careful stoichiometric control, solution-processed 2D perovskite films often exhibit disordered multiple-QW structures, which increase charge-carrier transport barriers^[10,11]. Low n -value phases ($n \leq 2$) can act as carrier recombination centers, causing non-radiative losses and affecting carrier extraction and optical absorption^[12]. Controlling QW distribution is crucial for improving device performance, with crystallization kinetics being key in determining QW distribution^[13-17]. Complex clusters in precursor solutions contribute to multiple-QW structures^[18,19]. Strategies such as additive engineering, component engineering, and process optimization have been employed to regulate QW distribution^[20-22]. For example, Peng et al. used acetic acid to reduce the polydispersity of the 2D perovskite phases at the early crystallization stage by coordinating AcOH and Pb^{2+} , thereby achieving a narrow QW distribution^[23]. Liang et al. replaced n -butylamine iodide (BAI) with n -butylamine acetate (BAAc) to form a uniform intermediate-phase gel via

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strong ionic coordination between BAAC and perovskite frameworks, resulting in a phase-pure 2D perovskite with $n = 4$ ^[24]. Zhang et al. employed a vacuum polarization method to induce forced nucleation, resulting in the uniform dispersion of 2D perovskites with different n values^[25]. Although these strategies effectively modulate QW distribution, the solvent's role in solution processing remains underexplored. Solvents influence the chemical environment and precursor interactions, shaping crystallization kinetics and QW distribution in 2D perovskites. Despite its importance, the impact of solvent engineering on QW distribution control has been insufficiently explored.

In this study, we introduce a solvent-hydrolysis-driven approach to fabricate single-QW 2D RPP films with adjustable QW widths ($n = 1-4$). Dimethylamine (DMA), produced from the hydrolysis of *N,N*-dimethylformamide (DMF), plays a crucial role in achieving a uniform single-QW structure by reducing complex clusters in the precursor solution, thereby ensuring uniform colloidal sizes and a homogeneous intermediate phase during crystallization. This method yields single-QW 2D perovskite photodetectors with exceptional performance, including a responsivity of 1153 mA/W, a detectivity of 6.98×10^{12} Jones, and excellent photoelectric imaging capabilities. These results offer valuable insights into solution-processed synthesis of single-QW 2D perovskites and provide a solid foundation for advancing high-performance 2D perovskite-based optoelectronic devices.

EXPERIMENTAL SECTION

Materials

Lead(II) iodide (PbI_2 , 99.999% metals basis), methylammonium iodide (MAI, $\geq 99\%$), dimethylammonium iodide (DMAI, 99.5%), propylammonium iodide (PAI, 99.5%), 2-Thiophenemethylammonium iodide (2-ThMAI, 99.5%), phenylmethylammonium iodide (PMAI, 99.5%) were purchased from Xi'an Yuri Solar Co., Ltd. Lead(II) diacetate ($\text{Pb}(\text{Ac})_2$, $\geq 99\%$), was purchased from Xi'an e-Light New Material Co., Ltd. Chlorobenzene (CB), dimethyl sulfoxide (DMSO), *N,N*-Dimethylformamide (DMF) were purchased from Sigma-Aldrich. Ammonium iodide (NH_4I , 99.999%) was purchased from MACKLIN, and Glasses were purchased from Yingkou OPV Tech New Energy Co., Ltd.

Synthesis of perovskite precursor solutions

PbI_2 -based precursor solutions: The precursor solutions were prepared in N_2 glovebox by mixing a stoichiometric amount of PbI_2 , MAI, and XI ($\text{X} = \text{PA}$, PMA, 2-ThMA) in DMF/DMSO (9:1) mixed solution, maintaining the concentration of Pb^{2+} at 0.1 mol/L and stirred at 40°C, 500 rpm for 2 hours.

$\text{Pb}(\text{Ac})_2$ -based precursor solutions: The precursor solutions were prepared in N_2 glovebox by mixing

stoichiometric amount of $\text{Pb}(\text{Ac})_2$, MAI, and XI ($\text{X} = \text{PA}$, PMA, 2-ThMA) in DMF/DMSO (9:1) mixed solution maintaining the concentration of Pb^{2+} at 0.1 mol/L with further addition of NH_4I at a molar ratio of 2:1 (with respect to Pb^{2+}) and stirred at 40°C (room temperature) or 100°C (high temperature), 500 rpm for 2 hours.

Device Fabrication

The glass substrates were cleaned in an ultrasonic cleaner for 60 minutes using a glass cleaner solution, deionized water, and acetone, followed by rinsing in an ethanol solution. The glass substrate was blown dry with a nitrogen gun and then treated with a plasma cleaner for 10 minutes before use. Then, the substrates were transferred into the glove box, and the perovskite precursor was spin-coated onto the substrates at 3000 rpm for 30 s. After 20 seconds, 250 μL of chlorobenzene was added as the anti-solvent, followed by annealing at 120 °C for 10 minutes to obtain the perovskite films. Lastly, a 50 nm thick layer of Au was evaporated onto the films. The channel length and width of the photodetectors were 17 μm and 64 μm , respectively.

Perovskite Film and Device Characterization

The X-ray diffraction (XRD) patterns were conducted using D2 PHASER XE-T XRD with Cu $K\alpha$ radiation (Bruker). The absorption spectra were measured using UV-vis absorption spectrophotometers (Hitachi UH-4150). The steady-state photoluminescence (PL) spectra and transient photoluminescence (TRPL) spectra were obtained using an FLS1000 Photoluminescence Spectrometer (Edinburgh Instruments Ltd.). The pH values were measured using a PHS-3C Acid Meter (INESA). The nuclear magnetic resonance (^1H NMR) spectra were measured using a Bruker 400MHz "AVANCE III" nuclear magnetic resonance system (NMR-400). AFM-IR measurements were carried out on an s-SNOM system (Attocube), the IR laser was set at 1000 cm^{-1} during the scanning. The morphology images of the perovskite films were measured by field emission SEM (QUATTRO S). The dynamic light scattering measurements were performed by a Zetasizer Nano ZS instrument (Malvern Instruments). The Fourier infrared (FT-IR) spectra were characterized using an FT-IR Spectrometer (PerkinElmer). The photodetection performance of the photodetectors was tested using an electrical probe stage (Everbeing C-4) and an Agilent 4155C semiconductor analyzer (Agilent Technologies, California, USA). A laser with a wavelength of 450 nm was used as the light source for the measurements, and the power of the incident irradiation was measured using a power meter (PM400, Thorlabs Inc.).

Calculation

All DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP). The interaction between the core and valence electrons was treated using the projected-augmented wave (PAW) method.

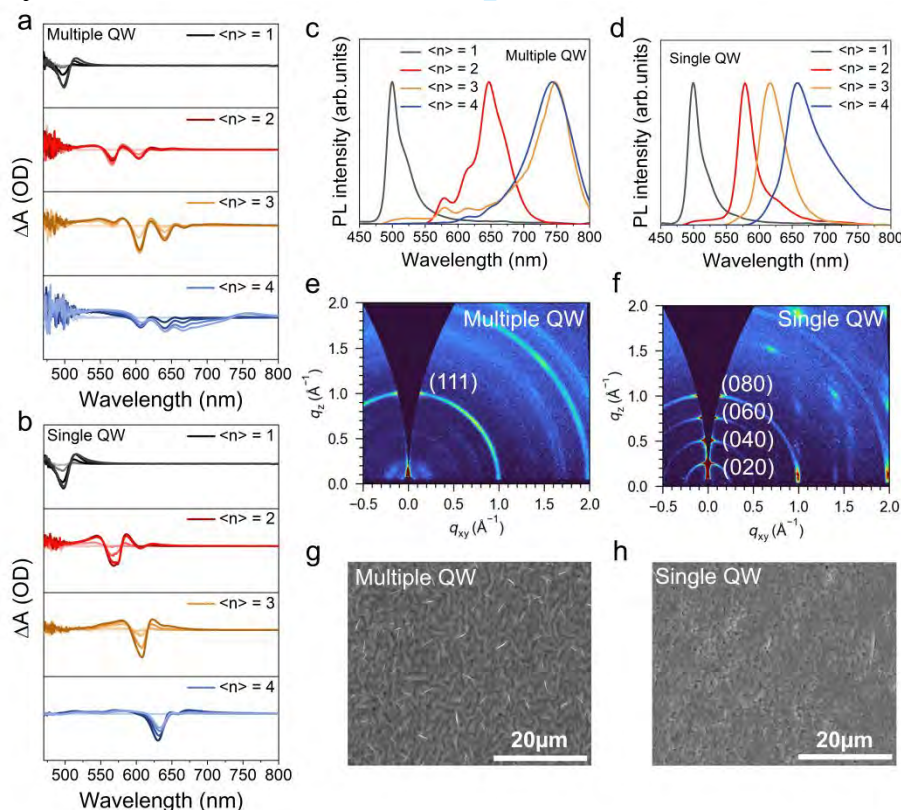
A kinetic energy cutoff of 400 eV was used for the planewave basis set for each compound. The exchange-correlation energy was evaluated by the Perdew-Burke-Ernzerhof (PBE) functional at the generalized gradient approximation (GGA) level. The energy tolerance of 1×10^{-7} eV was used in electronic structure calculations, and the geometry optimization was performed until all force was less than $0.05 \text{ eV } \text{\AA}^{-1}$. All structures are downloaded from the Material Project. A $(3 \times 3 \times 3)$ Monkhorst-Pack k-point grid was utilized to sample the Brillouin zone for the supercell model. The DFT+D3 method is used to account for dispersion interactions and correctly describe van der Waals forces. Formation energy is defined by the following equation: $E_{\text{form}} = E_{\text{AB}} - (E_{\text{A}} + E_{\text{B}})$.

RESULTS AND DISCUSSION

Characteristics of Multiple-QW and Single-QW Perovskite Films

We synthesized a series of single-QW RP perovskite films with the formula of $(\text{PA})_2\text{MA}_{n-1}\text{Pb}_n\text{I}_{3n+1}$ ($n = 1, 2, 3$, and 4) using a one-step solution method (Figure S1). Instead of conventional PbI_2 , we employed lead acetate ($\text{Pb}(\text{Ac})_2$) and ammonium iodide (NH_4I) as lead and halide sources, respectively, with precursor solutions stirred at elevated temperatures. The resulting films exhibited distinct colors, reflecting variations in band gaps influenced by quantum confinement (Figure S2). To investigate the QW distribution, we performed femtosecond transient

absorption (TA) spectroscopy. An ultrafast laser pulse excites the sample, generating charge carriers in the QWs, which then relax to their ground state or transfer between adjacent QWs via quantum coupling. These dynamics manifest as distinct photobleaching (PB) signals in the TA spectra, corresponding to excitonic transitions specific to each QW thickness (n -value) due to quantum confinement effects^[17,26]. Thus, the distribution of QW widths can be accurately determined by TA spectra. While precursor solutions were configured strictly according to stoichiometric ratios, the films prepared based on PbI_2 ($n = 2, 3, 4$) are still dominated by multiple PB peaks (Figure 1a), indicating a heterogeneous QW distribution. The peaks at 566, 604, 641, 665, and 750 nm correspond to band-edge exciton excitations in distinct QWs ($n = 2, 3, 4, 5$, and ∞ , respectively). In contrast, $\text{Pb}(\text{Ac})_2$ -based films (Figure 1b) exhibit a single dominant PB peak for all films, with negligible unwanted PB peaks, demonstrating a near-single QW structure. These results indicate that the $\text{Pb}(\text{Ac})_2$ -based approach effectively modulates QW distribution, achieving a near-single QW structure. UV-visible and steady-state photoluminescence (PL) tests further confirm these findings. The UV-visible pattern of multiple-QW films displays multiple exciton absorption peaks, whereas the single-QW films manifest a nearly single exciton absorption peak (Figure S3).



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Figure 1. TA spectra at different delay times of (a) multiple-QW and (b) single-QW films. PL spectra of (c) multiple-QW and (d) single-QW films. GIWAXS images of (e) multiple-QW and (f) single-QW films. Scanning electron microscopy image of (g) multiple-QW and (h) single-QW film.

As illustrated in Figures 1c and 1d, similar results are observed in the PL spectra. The multiple-QW films ($n > 2$) reveal complex multi-peak distribution, with low PL intensity for desired low n values and a constant shift in primary emission peaks towards the 3D phase (750 nm). This is attributed to the increasing 3D phase content with increasing n value and efficient charge carrier funneling from the thin to the thick QW^[27]. Conversely, single-QW films exhibit a distinct single peak, lacking the characteristic 3D phase emission peak. These results further demonstrate the method's ability to produce near-single QW structures.

X-ray diffraction (XRD) analysis elucidated phase compositions of multiple-QW and single-QW films (Figure S4). Multiple-QW films show diffraction periodic peaks structures of (00k) and (0k0) crystal planes for $n = 1$ and $n = 2$ samples. However, for $n \geq 3$, these periodic 2D structure peaks disappeared, replaced by diffraction peaks around 14° , indicative of multiple-QW and large n -value perovskite phases^[28]. In contrast, all single-QW films ($n = 2, 3, 4$) exhibit sharp diffraction peaks of periodic 2D crystal planes with narrower half-peak widths and higher diffraction peak intensities, suggesting purer QW distribution and

improved crystal quality. Grazing incidence wide-angle X-ray scattering (GIWAXS) measurements further analyzed crystal orientation (Figure 1e, f). The multiple-QW film exhibits diffraction rings with (111) planar signals, indicating significant polycrystalline randomness, which is consistent with the XRD results. Conversely, the single-QW film exhibits well-defined Bragg peaks, such as (020), (040), and (060), indicating superior crystallinity and orientation. Scanning electron microscopy (SEM) images (Figure 1g, h, and Figure S5) reveal a rugged and discontinuous morphology for multiple-QW films. In contrast, single-QW films exhibit a dense and well-grained morphology. Also, the TRPL results (Figure S6) reveal that the carrier lifetime of the perovskite film increases from 4.01 ns (multiple-QW) to 6.14 ns (single-QW) (the fitting data are presented in Table S1). The prolonged carrier lifetime suggests a lower density of defect states in the films. These characteristics, including single QW distribution, high crystallinity, and continuous morphology, are critical for optimizing the performance of 2D perovskite-based optoelectronic devices.

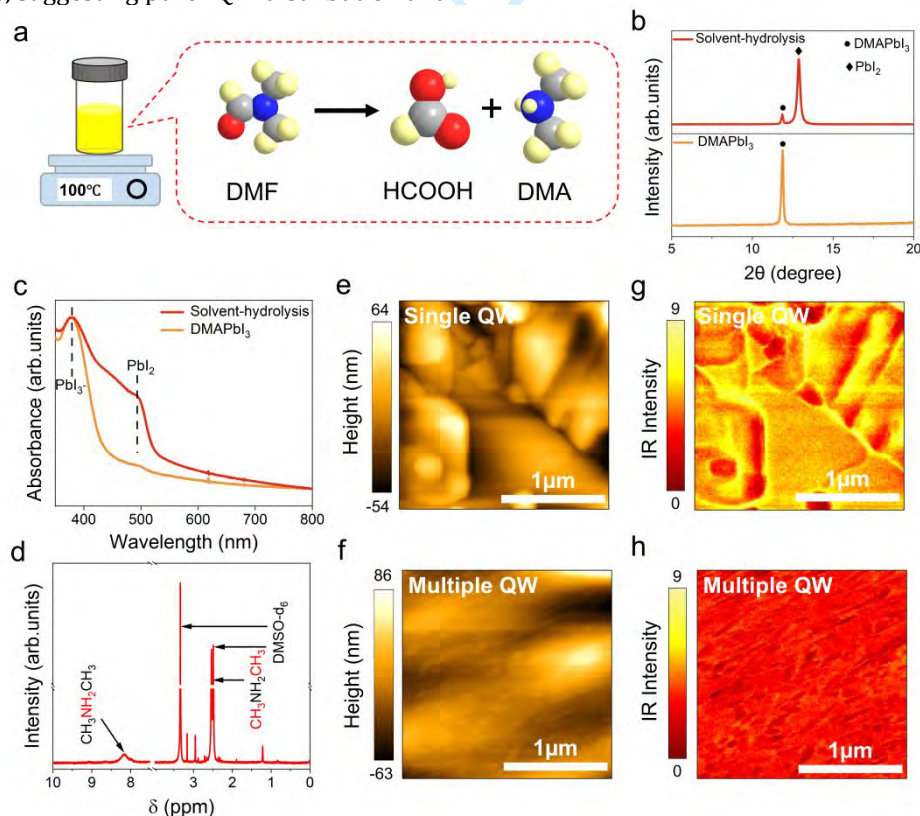


Figure 2. (a) Schematic of the hydrolysis of DMF to dimethylamine and formic acid. (b) XRD patterns, (c) UV-Visible absorption spectra, (d) ^1H NMR spectra of the solvent hydrolysis and DMAPbI₃ sample; the powder for NMR measurement was dissolved in DMSO-*d*₆. AFM topography images of (e) single-QW and (f) multiple-QW films. AFM-IR images of (g) single-QW and (h) multiple-QW films illuminated at 1000 cm^{-1} .

Mechanism for Engineering Single-QW Structures

Variations in composition significantly influence the physicochemical properties of precursor solutions, leading to notable differences in the resultant film^[29]. Using $\text{Pb}(\text{Ac})_2$ and NH_4I as precursors generates ammonium acetate (NH_4Ac), which decomposes rapidly at elevated temperatures to yield acetic acid^[30,31]. This decomposition is confirmed by pH measurements (Figure S7). The liberated acetic acid catalyzes the hydrolysis of the solvent DMF to dimethylamine (DMA)^[32], as depicted in Figure 2a. The generation of DMA is corroborated by XRD, UV-visible, and nuclear magnetic resonance (NMR) analyses. Specifically, XRD patterns of films prepared with $\text{Pb}(\text{Ac})_2$ and NH_4I exhibit a characteristic DMAPbI₃ peak at 11.8° (Figure 2b), while UV-visible absorption spectra reveal distinct peaks corresponding to DMAPbI₃ (Figure 2c)^[33]. Furthermore, the ^1H NMR spectra (Figure 2d) display diagnostic signals at $\delta = 8.15$ ppm and $\delta = 2.54$ ppm, corresponding to the protons in the $-\text{NH}_2^+$ and $-\text{CH}_3$ groups, respectively^[34]. The integration ratio of these signals (corresponding to the molar ratio) is 1:3,

aligning with the chemical formula DMA ((CH_3)₂ NH_2) (Figure S8). High-resolution AFM-IR spectroscopy was employed to characterize the presence of DMA in multiple-QW and single-QW films. AFM images (Figure 2e, f) reveal that single-QW films exhibit larger, more well-defined grains than multiple-QW films, consistent with prior SEM results. AFM-IR mapping at 1000 cm^{-1} (Figure 2g, h), corresponding to the stretching vibration $\nu_{\text{C-N}}$ (DMA) (Figure S9, S10)^[35], reveals a uniform DMA signal in the single-QW film. In contrast, the multiple-QW film shows a negligible signal, confirming the presence of DMA in single-QW films. In the FTIR spectrum of the $\text{Pb}(\text{Ac})_2$ -based film (Figure S11), none of the C=O stretching vibration peaks corresponding to the acetate ion were observed. This finding suggests that acetate-related products are not present in the final product. To demonstrate DMA is responsible for creating single-QW structures, a solution of $\text{Pb}(\text{Ac})_2$ and NH_4I was stirred at room temperature, yielding a neutral solution (Figure S12). Films prepared from this solution show no DMA signals in XRD, UV-visible, or NMR analyses (Figure S13-15). Moreover, the UV absorption of perovskite

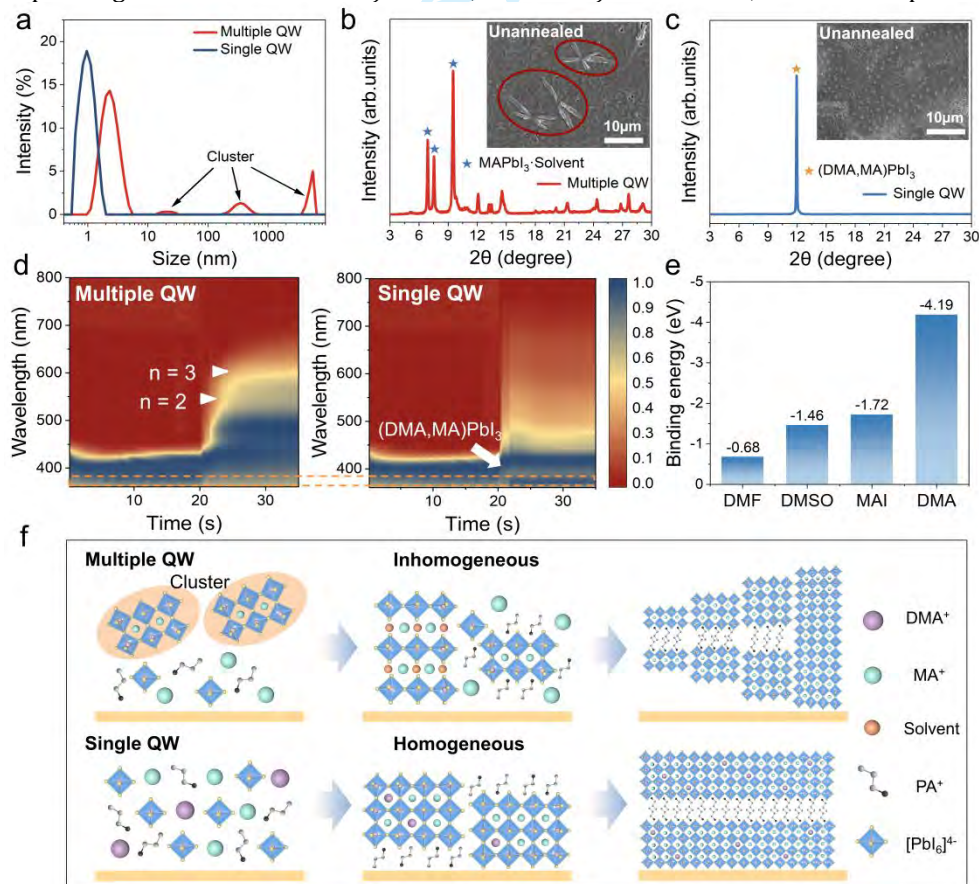


Figure 3. (a) Particle size distributions in the multiple-QW and single-QW precursor solutions. XRD patterns of (b) unannealed

multiple-QW film and (c) unannealed single-QW film. The inset figure shows the SEM image. (d) In situ UV-Visible absorption during spinning of multiple-QW and single-QW films. (e) Calculated binding energies between PbI_2 and DMF/DMSO/MAI/DMA, respectively. (f) Schematic diagram of the crystallization process for multiple-QW and single-QW films.

films prepared under room temperature stirring and incorporating acetic acid as an additive both exhibit multiple exciton absorption peaks (Figures S16 and S17). Conversely, films prepared by incorporating DMAI into the PbI_2 -based system display a single exciton absorption peak in their ultraviolet absorption spectra (Figure S18). These results conclusively demonstrate that DMA is essential for forming single-QW perovskite films.

Introducing DMA into the precursor solution significantly affects the crystallization dynamics of the perovskite film, including solution formation, gelation, and nucleation^[36,37]. Prior studies have established that perovskite precursors typically comprise complex colloidal particle mixtures with random size distributions, serving as nucleation centers and forming QWs of varying widths^[38]. To investigate this, we employed dynamic light scattering (DLS) to analyze particle size distributions in PbI_2 -based and $\text{Pb}(\text{Ac})_2$ -based precursor solutions (Pb^{2+} concentration of 0.1 mol/L) at room temperature (Figure 3a). The PbI_2 -based solution exhibited larger particles with pronounced clustering, whereas the $\text{Pb}(\text{Ac})_2$ -based solution displayed smaller, more uniform particles. This uniform colloid size distribution is crucial for achieving a single-QW structure, as variations in colloid size result in distinct intermediate phases during crystallization^[39]. Optical images of two unannealed films (Figure S19) reveal marked color variations, suggesting substantial differences in their intermediate phases. Unannealed multiple-QW films show solute-solvent characteristic XRD peaks and the needle-like complexes observed in the SEM morphology (Figure 3b)^[40], suggesting an inhomogeneous intermediate phase that likely results in multiple-QW structures^[19,26]. In contrast, the presence of DMA in the single-QW film promotes the formation of a DMA-PbI_3 -like intermediate phase (Figure 3c). XRD patterns of the unannealed single-QW film exhibit a right-shift compared to the DMA-PbI_3 film (Figure S20), indicating partial replacement of DMA by MA, leading to a lattice contraction and formation of a homogeneous $(\text{DMA,MA})\text{PbI}_3$ intermediate phase. The SEM morphology of unannealed single-QW films also reveals a more homogeneous morphology. In situ absorption spectroscopy (Figure 3d) provides further evidence of the difference in the intermediate phases, with multiple-QW film exhibiting absorption peaks with various n values following anti-solvent addition, while the single-QW films primarily display $(\text{DMA,MA})\text{PbI}_3$ absorption peaks. These results from DLS, XRD, and in situ absorption analyses collectively demonstrate that DMA effectively suppresses the formation of complex mixtures, thereby fostering a uniform intermediate phase. Density functional theory

(DFT) calculations (Figures 3e and S21) further validate that DMA stabilizes a distinct intermediate phase, thereby inhibiting the formation of complex mixtures. The binding energy between DMA and PbI_2 (-4.19 eV) is lower than that of PbI_2 and DMF (-0.68 eV), DMSO (-1.46 eV), and MAI (-1.72 eV), indicating a preferential reaction of PbI_2 with DMA. Based on the analysis above, a schematic illustration (Figure 3f) elucidates how DMA modulation ensures uniform particle size distribution, fostering homogeneous intermediate phases and enabling the formation of a single-QW structure.

To evaluate the versatility of this strategy, we employed two different types of spacer cations, phenylmethylamine (PMA) (benzene-type) and 2-thiophenemethylammonium (ThMA) (thiophene-type), for fabricating 2D perovskite films. In Figure S22, all the UV-visible absorption spectra of $(\text{PMA})_2\text{MA}_{n-1}\text{Pb}_n\text{I}_{3n+1}$ and $(\text{ThMA})_2\text{MA}_{n-1}\text{Pb}_n\text{I}_{3n+1}$ ($n = 1, 2, 3, 4$) perovskite films exhibit a single absorption peak, confirming the method's effectiveness in producing single-QW 2D perovskite structures across diverse spacer cations.

Performance Evaluation of the Perovskite Photodetector

To assess the optoelectronic properties of multiple-QW and single-QW perovskite films, we fabricated photodetectors with a glass/perovskite/Au architecture (Figure S23). Photodetector performance was evaluated using a 450 nm laser source, with on/off current-time characteristics measured at a light intensity of 39.2 mW/cm² and a bias voltage of 1.5 V. The dynamic photoresponse, shown in Figure 4a, demonstrates stable and reversible photoswitching across all devices. Notably, the single-QW device exhibits a photocurrent approximately tenfold higher than the multiple-QW device, which is attributed to its superior film quality and enhanced charge transport properties. Current-voltage (I - V) characteristics under varying light intensities (Figures 3b, c) reveal a progressive increase in photocurrent with increasing light intensity. The single-QW device achieves an on/off ratio of 5000, significantly outperforming the multiple-QW device. The photocurrents at varying light intensities (Figure S24a) show a standard linear relationship between photocurrent and light intensity. Key performance metrics, including responsivity (R), detectivity (D^*), and external quantum efficiency (EQE), were calculated to assess the photodetectors' light detection capabilities^[41,42]. R , defined as $R = I_p / \Phi S$, where I_p is the photocurrent ($I_p = I_{\text{light}} - I_{\text{dark}}$), Φ is the light intensity, and S is the photodetector's active area, quantifies the efficiency of the photodetector's response to optical signals. As shown in Figure 4d, the single-QW device demonstrates superior responsivity across all tested light intensities,

achieving a maximum R of 1153 mA/W at 0.00392 mW/cm², compared to 22 mA/W. The D^* , indicative of the photodetector's sensitivity, is calculated as $D^* = S^{1/2}/\text{NEP}$, where the noise equivalent power (NEP) is given by $\text{NEP} = I_n/R$, and I_n is the current noise spectral density (Figure S25). At 0.00392 mW/cm², the single-QW device yields a D^* of 6.98×10^{12} Jones, markedly higher than the multiple-QW device's D^* of 2.66×10^{10} Jones (Figure 4e). The EQE, which reflects the efficiency of photon-to-electron conversion, reaches 353% for the single-QW device, compared to 7% for the multiple-QW device (Figure S24b). These enhancements in the single-QW device are driven by improved crystalline quality, reduced non-radiative recombination, and superior carrier transport. Additionally, the single-QW device exhibits rapid response times of 427 μs (rise time) and 445 μs

(decay time), as shown in Figure 4f. As the photoresponse of detectors varies with wavelength, the spectral response characteristics of single-QW and multiple-QW devices were measured across the 400–800 nm (Figure S26). These detectors predominantly demonstrate a photoresponse within the 400–700 nm range. However, due to the more uniform QW distribution in single-QW devices, superior detection capabilities within specific bands are demonstrated, exhibiting virtually no response at 760 nm (corresponding to the 3D phase). Conversely, multiple-QW devices demonstrate responses at multiple wavelengths, suggesting a more disordered QW distribution. A comparison with other 2D perovskite-based photodetectors (Table S2) showcases the exceptional performance of the single-QW device.

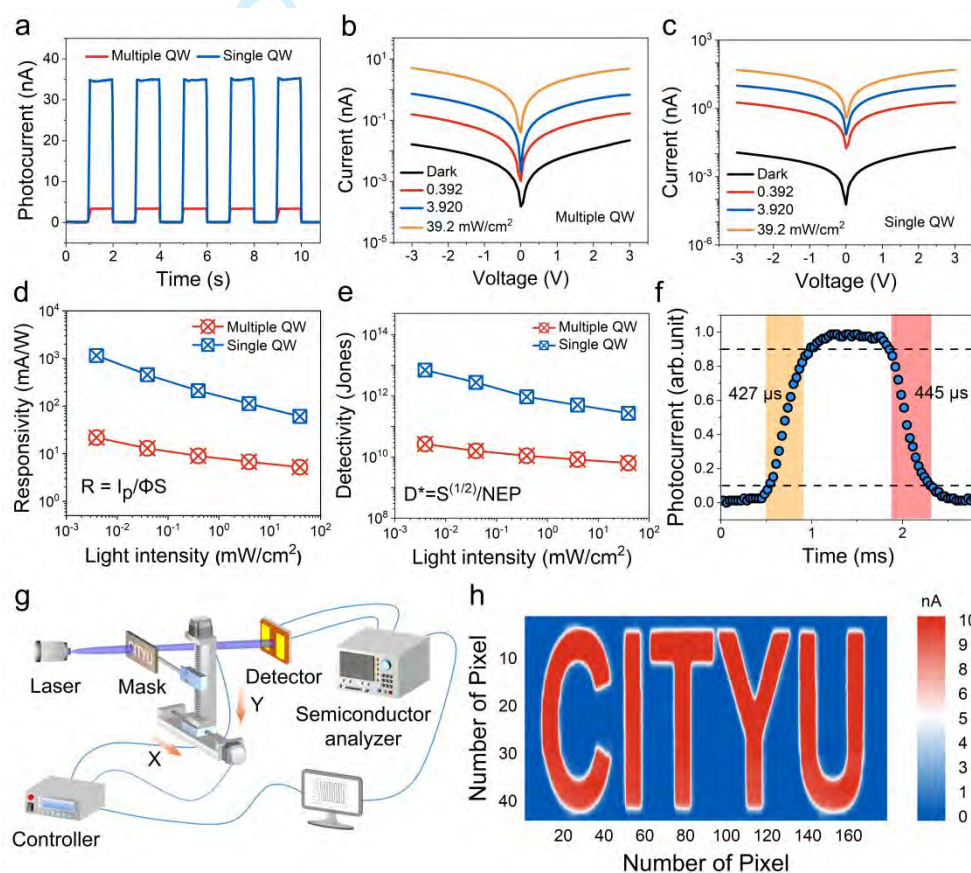


Figure 4. (a) Time-dependent on-off switching of the perovskite devices prepared from multiple-QW and single-QW perovskite films. I-V curves of photodetector devices prepared from (b) multiple-QW perovskite film and (c) single-QW perovskite film. Dependence of (d) responsivity, (e) detectivity on the light intensity of the perovskite devices prepared from multiple-QW and single-QW perovskite films. (f) High-resolution current-time curve for the single-QW perovskite photodetector. (g) Schematic diagram of the imaging system. (h) Current mapping imaging of "CITYU" was obtained from the single-QW device testing at a 450 nm laser.

Given the superior photosensitivity of the single-QW device, we further explored its potential for image sensing applications. Figure 4g schematically illustrates the photoelectric imaging system, which employs a 450nm laser light source for the

experimental procedure. A displacement stage controls the movement of the mask plate, enabling the light source to scan the pattern. The semiconductor analyzer records the light response at various positions, producing a high-resolution "CITYU" pattern

(Figure 4h). In contrast, the multiple-QW device generates a "CITYU" image with poor resolution (Figure S27). These results, along with the device's excellent optoelectronic performance, highlight its significant potential for high-performance optoelectronic applications.

Furthermore, the poor stability of perovskite films remains a major obstacle to their commercialization. In 2D perovskite, long organic cation spacers are believed to inhibit reactions with water and oxygen, thereby enhancing stability^[43]. However, in multiple-QW films, 3D and large n-value phases typically form at the top of the film, compromising its stability^[44]. Therefore, we evaluated the stability of multiple-QW and single-QW samples under thermal, humidity, and light conditions. Thermal stability was evaluated by continuous heating at 100 °C. XRD patterns, shown in Figure 5a and 5b, reveal that after 72 hours, a characteristic PbI_2 peak at 12.8° ^[45] emerges in the multiple-QW film, indicating degradation. In contrast, the single-QW film exhibits

no changes in its XRD peaks even after 120 hours, demonstrating superior thermal stability. Humidity stability was tested by storing both films in a controlled environment at 85% relative humidity. As shown in Figure 5c, the characteristic XRD diffraction peaks of the multiple-QW film nearly disappear after 24 hours, signifying significant degradation. Conversely, the single-QW film's diffraction peaks remain unchanged after 96 hours (Figure 5d), highlighting its exceptional resistance to humidity-induced degradation. Light stability was examined under 39.2 mW/cm^2 illumination and 1.5 V bias for 5000 s in an ambient environment (Figure 5e, f). The photocurrent of the multiple-QW device drops to approximately 10% of its initial value after 5000 s. In contrast, the single-QW device retains about 85% of its initial photocurrent, underscoring its remarkable photostability. The enhanced stability of single-QW devices is ascribed to the uniform spacer layer protection and the improved structural integrity of the perovskite film.

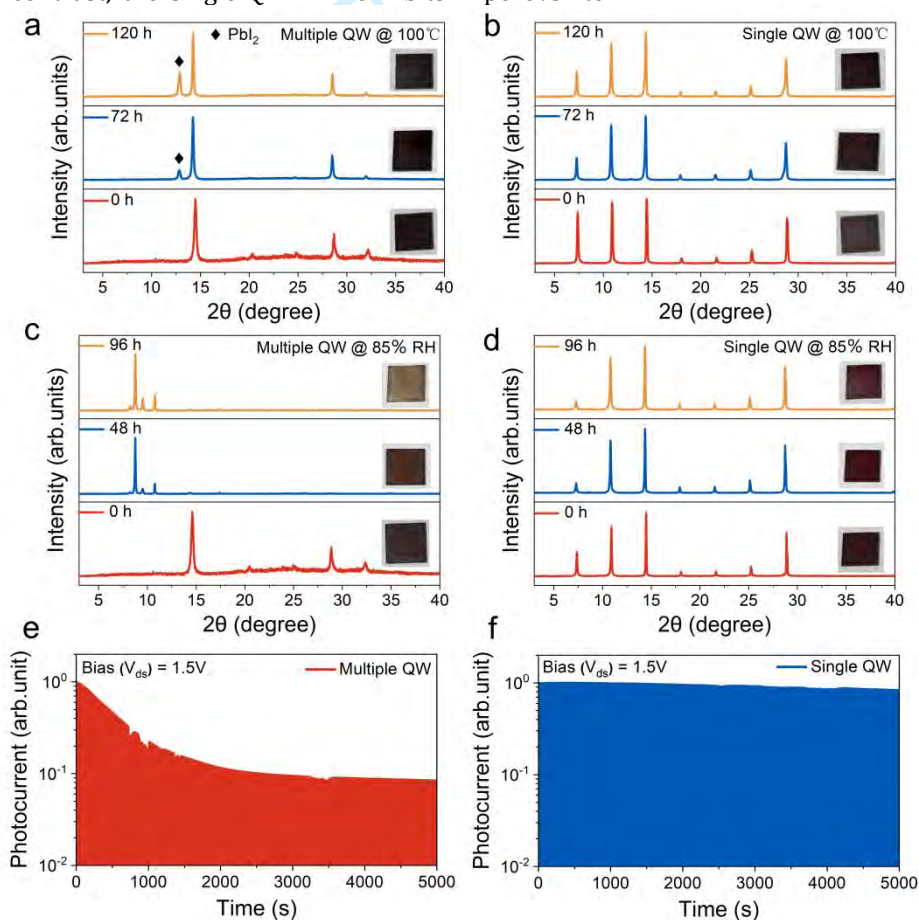


Figure 5. XRD patterns of multiple-QW and single-QW perovskite films exposed in (a, b) under 100 °C heating in N_2 , (c, d) ambient air with RH of 85% at room temperature, (e, f) under continuous light (39.2 mW/cm^2) soaking in ambient air.

CONCLUSIONS

In summary, our study highlights the crucial role of

DMA, formed through DMF hydrolysis, in the precise synthesis of 2D perovskite films with consistent

single-QW distribution. This method overcomes the challenges associated with traditional deposition techniques. By reducing cluster formation in the precursor solution, DMA ensures a uniform colloidal particle size, thereby preventing the development of solute-solvent intermediate phases that typically result in multiple-QW structures during early crystallization. The single-QW 2D perovskite films were produced and validated with various organic ligands, demonstrating broad applicability and consistent reproducibility. Photodetectors made from these films show exceptional performance, including a responsivity of 1153 mA/W, a detectivity of 6.98×10^{12} Jones, and remarkable stability under thermal, humidity, and light exposure. Furthermore, successfully integrating these single-QW films into high-resolution visible image sensors highlights their superior imaging capabilities. This work establishes a solid foundation for the strategic design and production of single-QW 2D perovskites, offering great potential for future optoelectronic devices.

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Conflict of Interest: The authors declare that they have no conflict of interest.

Supplementary information Experimental details and supporting data are available in the online version of the paper.



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溶剂水解驱动有序单量子阱二维钙钛矿工程

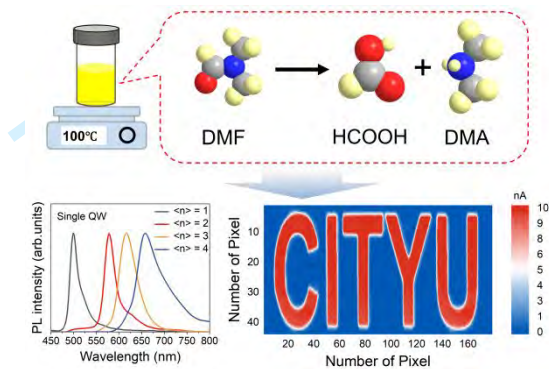
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单量子阱二维钙钛矿材料凭借其卓越特性和稳定性, 有望革新光电子器件。然而, 溶液加工的二维钙钛矿常形成无序的多量子阱结构, 导致器件性能差。本文提出一种溶剂水解驱动的结晶动力学调控方法, 成功制备出高度有序的单量子阱二维钙钛矿薄膜。N,N-二甲基甲酰胺水解产物二甲胺作为关键介质, 抑制前驱体溶液中胶体的团聚, 形成均匀中间相, 使得高度有序的单量子阱薄膜能够从中间相中结晶生长。基于单量子阱薄膜所制备的光探测器展现出1153 mA/W的响应度以及 6.98×10^{12} Jones的探测率的卓越性能, 并在环境条件下具有优异的稳定性。我们的研究为高性能二维钙钛矿材料提供了可扩展的溶液加工策略, 为先进光电子应用开辟了新前景。

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Dimethylammonium (DMA), derived from the hydrolysis of N,N-dimethylformamide (DMF), a solvent, effectively suppresses the formation of complex clusters in the precursor solution. This inhibits the development of a solute–solvent intermediate phase during crystallization, enabling the formation of single quantum well 2D perovskite structures. The resulting photodetectors demonstrate excellent performance and stability.

Keywords: 2D perovskite, Single-quantum-well, Solvent hydrolysis, Optoelectronic device, Stability



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